

Topological Quantum Computation

An Odyssey

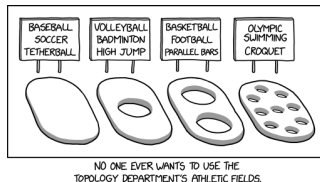
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What is topological quantum computation?

Topology: the study of properties that are resistant to deformation.



Problem: Quantum computers are very sensitive.

Solution: Build a quantum computer using topological properties!

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What are particles?

QFT tells us particles are just excitations in a field.

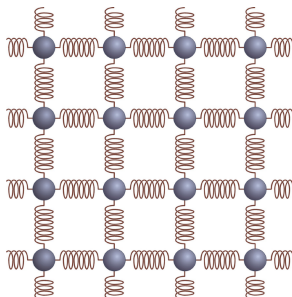
$$a = \frac{1}{\sqrt{2}}(q + ip) = \frac{1}{\sqrt{2}} \left(q + \frac{d}{dq} \right)$$

$$a^\dagger = \frac{1}{\sqrt{2}}(q - ip) = \frac{1}{\sqrt{2}} \left(q - \frac{d}{dq} \right).$$

Energy gap = mass of the particle

What are quasiparticles?

Any system with local excitations can be described as having *quasiparticles*.

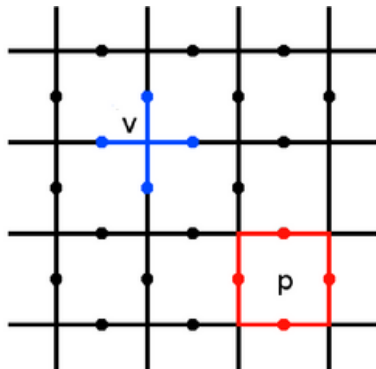


Lattice with vibrational modes $\implies$ *phonons* (quantized sound waves)

The Toric Code

System of qubits on the *edges* of a lattice.

Identify opposite edges of lattice $\implies$ torus!



The Plaquette Operators

Can move between states using *plaquette operators*

$$A_s = \prod_{\substack{\text{edges } i \\ \text{touching} \\ \text{vertex } s}} \sigma_i^z \qquad B_p = \prod_{\substack{\text{edges } i \\ \text{touching} \\ \text{square } p}} \sigma_i^x$$

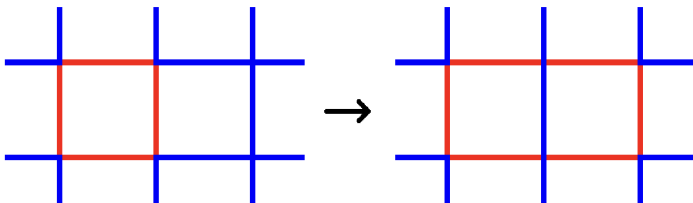


Figure: Map between eigenstates of A_s using B_p

The Ground State of the Toric Code

$$H = - \prod_s A_s - \prod_p B_p$$

“Derivation” of ground state

- Any set of closed loops is a ground state of $-\prod_s A_s$
- $\prod_p B_p$ moves between loops
 $\implies$ ground state must be indifferent to any B_p .
- Ground state must be an *equal-weight superposition* of all closed loops reachable by B_p

The Topology of the Toric Code

Which loops can be transformed into each other by B_p ?

- In the plane: all of them (1 equivalence class).
- On the torus: 4 equivalence classes

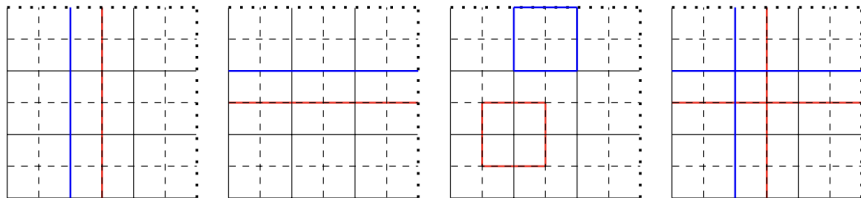


Figure: Equivalence classes of loops on the torus

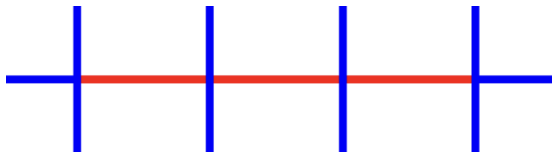
Can use cleanly separated ground states to build an error correction scheme. (Wang, Alice (2024). *The Toric Code*)

Excitations in the Toric Code

We can create localized energy excitations using a *string operator*.

$$S = \prod_{i \in \text{string}} \sigma_i^x$$

These are called *electric charges* (for some reason).



We can move it around with σ_i^x

Is that the only kind?

Duality in the Toric Code

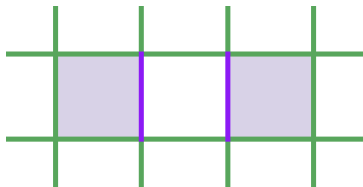
H doesn't really care about σ^z or σ^x eigenstates.

Let's consider a lattice filled with B_p (σ_i^x) eigenstates.

Now A_s transforms loops in the *dual lattice*.

Can create excitations in the dual lattice using the string operator

$$\tilde{S} = \prod_{i \in \text{dual string}} \sigma_i^z$$



These are *magnetic vortices*.

Particle Statistics

We have two quasiparticles: e and m . What can we know about them?

Definition

The *statistics* $R(a, b)$ of a pair of (quasi)particles (a, b) is the phase change acquired by the exchange of the two particles.

Definition

The *exchange statistics* $R(a, a)$ of a (quasi)particle a is the statistics of the particle with itself.

Kawagoe & Levin: Method to compute the statistics of quasiparticles.

Statistics of “real particles”

All “real particles” fall into one of two categories

Definition

A *boson* is a particle with exchange statistics of $1 = e^{0i}$

Definition

A *fermion* is a particle with exchange statistics of $-1 = e^{\pi i}$

Theorem (Spin-Statistics Theorem)

- Every “real particle” is either a boson or a fermion
- The statistics are determined entirely by the spin of the particle
- The fusion of bosons and fermions is also determined

Fusion of “real particles”

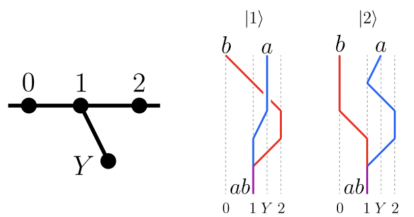
Can bring two particles together for new statistics.

- The fusion of two fermions is a boson (Cooper pairs)
- The fusion of two bosons is a boson

Can be derived by considering spin

Statistics in the Toric Code

For a general lattice model, define the statistics of quasiparticles by the following procedure



$$|1\rangle = M_{01}^b M_{12}^b M_{Y1}^a S(a, b) |ab_1\rangle$$

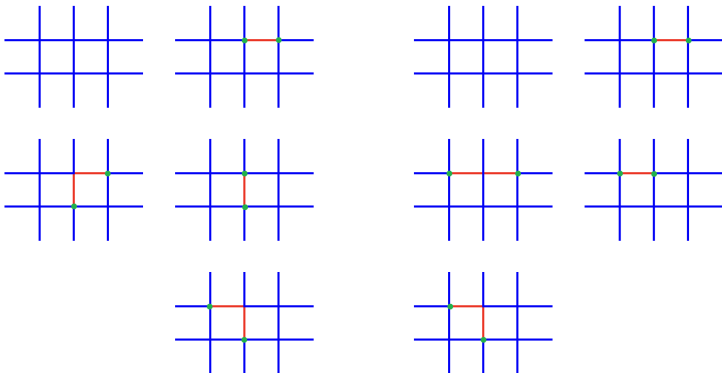
$$|2\rangle = M_{Y1}^a M_{12}^a M_{01}^b S(b, a) |ab_1\rangle$$

$$R(a, b) = \langle 2 | 1 \rangle$$

Statistics of e, m

Note that $e + e = 1$, so our initial state is the “vacuum”.

$$S(e, e) = \sigma^x \quad M_{ij}^e = \sigma^x$$

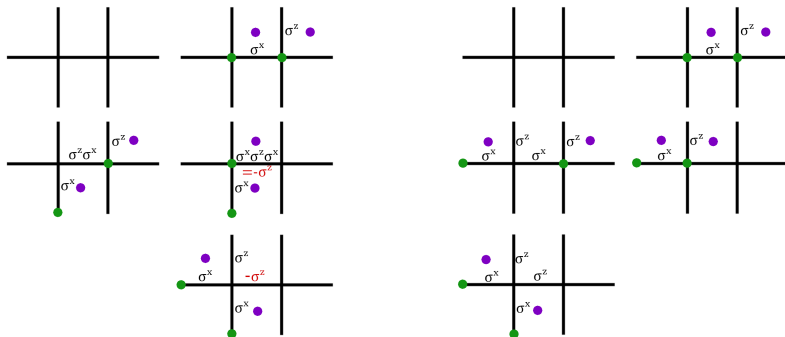


States are identical $\implies R(e, e) = 1$. Same with m .

Statistics of *em* fusion

We can move *em* just like we move *e* and *m* individually

$$S(em, em) = \sigma_{i'}^z \sigma_i^x \quad M_{ij}^e = \sigma_{i'}^z \sigma_i^x$$



$R(em, em) = \langle 2|1 \rangle = -1?!?! \text{ What's going on here?}$

Anyons

Definition

An *anyon* is quasiparticle that is neither a boson nor a fermion.

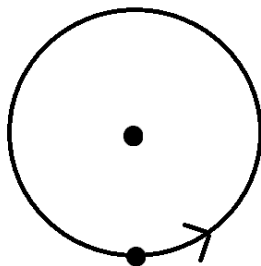
Definition

An *anyon theory* is a way of characterizing the anyons in a system.

1. **Set of anyon types:** a finite set of anyon types, $\mathcal{A} = \{a, b, c, \dots\}$.
2. **Fusion product:** an associative and commutative multiplication law on $\mathcal{A}$, denoted $a \times b$ or ab .
3. **F -symbol:** a function $F : \mathcal{A} \times \mathcal{A} \times \mathcal{A} \rightarrow U(1)$, denoted $F(a, b, c)$.
4. **R -symbol:** a function $R : \mathcal{A} \times \mathcal{A} \rightarrow U(1)$, denoted $R(a, b)$.

The Hole in the Spin-Statistics Theorem

Performing the exchange twice looks like this:



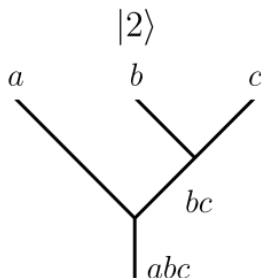
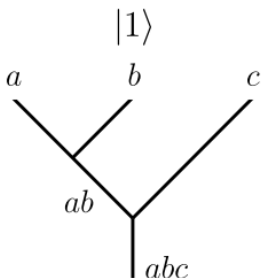
In 3D: contracts to identity $\implies R(a, a) = \pm 1$

In 2D: non-contractable $\implies R(a, a)$ can be *anything*!

Fusion of anyons

We've seen that the fusion of anyons can be wild.

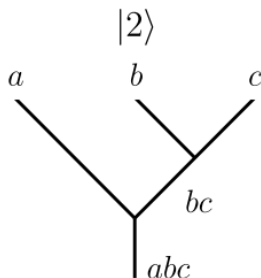
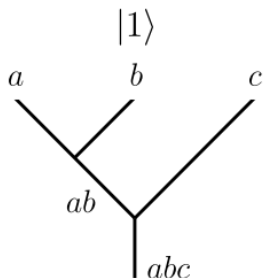
Q: Are these two ways of fusing anyons the same?



Fusion of anyons

We've seen that the fusion of anyons can be wild.

Q: Are these two ways of fusing anyons the same?



A: In general, no. $F(a, b, c) = \langle 2|1\rangle$. Kawagoe & Levin compute F as well.

Fusion outcomes

How wild can fusion get? Consider the fusion of fermions

$$e \otimes e = \{\text{spin-1 "triplet"}\} \oplus \{\text{spin-0 "singlet"}\}$$

Triplet and singlet states have the same statistics $\implies$ boring

What if quasiparticles could fuse into different anyons depending on their state?

Abelian and non-abelian anyons

Definition

An anyon is *abelian* if it always has exactly one fusion outcome.

Definition

An anyon is *non-abelian* if it can have multiple fusion outcomes.

Experimental history of anyons

- 1977: Anyons first theorized
- 1982: Fractional Quantum Hall Effect discovered
 - ▶ Anyons used for theoretical description

Experimental history of anyons

- 1977: Anyons first theorized
- 1982: Fractional Quantum Hall Effect discovered
 - ▶ Anyons used for theoretical description

(40 years later)

- 2020: First definitive detection of anyons
- 2022: Detection and braiding of non-abelian (Ising) anyons (Google)

Fibonacci anyons

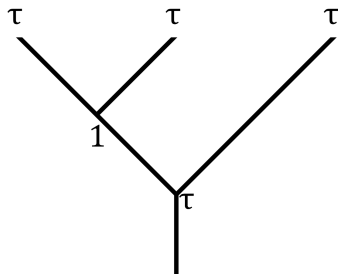
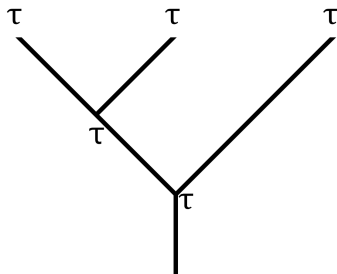
Anyons $\{1, \tau\}$ defined by the following fusion rule

$$\tau \times \tau = 1 + \tau$$

Then

$$\tau \times \tau \times \tau = 2 \cdot \tau + 1$$

i.e. 2 pathways to fuse $|\tau\tau\tau\rangle$ to $|\tau\rangle$

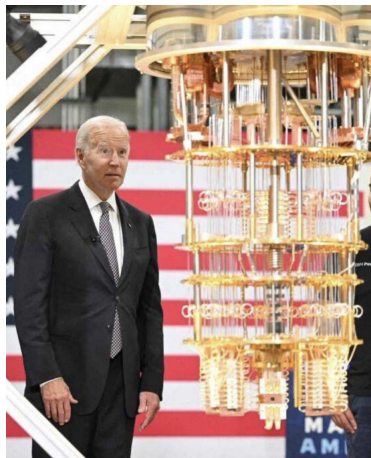


Mutual exclusivity $\implies$ orthogonal states
 $\implies$ basis for qubit!

Where are the qubits?

- Photonic: state of photon
- Superconducting: ground and excited states
- Topological: ???

Convince yourself that
 $|\tau\tau\tau; 1\tau; \tau\rangle$ and $|\tau\tau\tau; \tau\tau; \tau\rangle$
are genuine basis states



F and R

F and R get promoted to *unitary matrices*²

$$F = F_{\tau\tau\tau}^{\tau} = \begin{pmatrix} \phi^{-1} & \phi^{-1/2} \\ \phi^{-1/2} & -\phi^{-1} \end{pmatrix}, \quad R = \begin{pmatrix} R_{\tau\tau}^1 & 0 \\ 0 & R_{\tau\tau}^{\tau} \end{pmatrix} = \begin{pmatrix} e^{\frac{4\pi i}{5}} & 0 \\ 0 & e^{\frac{-3\pi i}{5}} \end{pmatrix}$$

Can be derived from the fusion rule using certain consistency conditions (*pentagon* and *hexagon* identities).

- F is allowed to mix fusion states.
- R is allowed to result in different phases after fusion.

²Ville, L., Pachos, J. *A Short Introduction to Topological Quantum Computation* 

Braid Operations

$$\begin{array}{c}
 \begin{array}{ccc}
 b & a & c \\
 & \searrow & \nearrow \\
 & e & \\
 & | & \\
 & d &
 \end{array}
 & = R_{ab}^e &
 \begin{array}{ccc}
 a & b & c \\
 & \searrow & \nearrow \\
 & e & \\
 & | & \\
 & d &
 \end{array}
 \end{array}$$

$$\begin{array}{c}
 \begin{array}{ccc}
 a & c & b \\
 & \searrow & \nearrow \\
 & e & \\
 & | & \\
 & d &
 \end{array}
 & = (F_{abc}^d)^{-1} R (F_{abc}^d) &
 \begin{array}{ccc}
 a & b & c \\
 & \searrow & \nearrow \\
 & e & \\
 & | & \\
 & d &
 \end{array}
 \end{array}$$

Surprising result: for Fibonacci anyons, R and $F^{-1}RF$ form a dense subset of $SU(2) \implies$ we can get any quantum algorithm just by braiding!

How to Compute

- Initialize n anyons
- Braid the anyons to perform unitary operations on the fusion basis
- Measure states by measuring the fusion channel

The point:

- Braiding is *topological*
- The basis qubit states are *protected*
- We can get *incredibly* error-resistant qubits and gates

When will this be realized experimentally? I don't know.